

Question 3:

3(a)

```
#include <iostream>
```

```
void f(int y) {  
    int x = y + 1;  
    for (int i = 0; i < 3; i++) {  
        int temp = i * 2 + x;  
        std::cout << i << " " << temp << " " << x << std::endl;  
    }  
}
```

```
int main(int argc, char *argv[]) {  
    int x = 2;  
    f(x);  
    std::cout << x << std::endl;  
    return 0;  
}
```

OUTLINE:

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└ GLOBAL SCOPE

└─ f

└ main

└─ argc

└─ argv

└─ x = 2

└ f(x)

└─ y = 2

└─ x = 3

└─ for-loop (Iteration 0)

| └─ i = 0

| └ temp = 3

└─ for-loop (Iteration 1)

| └─ i = 1

| └ temp = 5

└─ for-loop (Iteration 2)

└─ i = 2

└ temp = 7

3(b)

```
#include <iostream>

int main() {
    int a = 5;

    int &ref_To_An_A = a;

    {
        int a = 10;

        std::cout << ref << std::endl;
    }

    return 0;
}
```

OUTLINE:

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└ GLOBAL SCOPE

└ main

└─ a = 5

└─ ref_To_An_A (refers to outer a)

└ Inner Block Scope

└ a = 10 (shadows outer a)

3(c & d) ### Unless I have lost my mind, these are the same code block.

```
-----  
#include <iostream>  
  
int main() {  
    int* p = new int(42);  
  
    {  
        int* q = p;  
        std::cout << *q << std::endl;  
    }  
    delete p;  
}
```

OUTLINE:

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└ GLOBAL SCOPE

└ main

└─ p (int* stack pointer)

└─ Inner Block Scope

| └ q (int* stack pointer -> points to p heap memory)

└ HEAP STORAGE

└ int(42) (Dynamic Memory)

└ delete p